



AI Developer

AI Career Roadmap Assignment
for Cohort 2

PRESENTATION BY:
DRUELL ALSTROM

What is an AI Developer ?

- An **AI developer** is a professional who designs, builds, and implements applications powered by artificial intelligence.

Why this role ?

- The role of an AI developer appeals to me because it blends creativity with technical problem solving. It also offers the chance to build impactful systems that learn and adapt, improving real-world applications across industries.





Key Responsibilities

- **Application Development:** Create AI-powered apps (chatbots, recommendation engines, fraud detection systems).
- **Integration:** Embed ML/NLP models into existing software platforms.
- **Collaboration:** Work with data scientists, ML engineers, and product managers to deliver AI features.
- **Optimization:** Ensure AI systems run efficiently and scale with user demand.
- **Maintenance:** Monitor AI performance and update models as data evolves.

Essential Skills

- **Programming:** Python, Java, C++, or JavaScript.
- **Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn.
- **APIs & Tools:** REST APIs, cloud services (AWS, Azure, GCP).
- **Databases:** SQL, NoSQL for handling structured/unstructured data.
- **Applied AI:** Natural Language Processing (NLP), Computer Vision, Recommendation Systems.

Tools & Technologies

Programming Languages

- **Python** – dominant in AI for libraries like TensorFlow, PyTorch, Scikit-learn.
- **Java & C++** – used for performance-heavy applications.
- **JavaScript** – for AI in web apps (e.g., TensorFlow.js).

Machine Learning & Deep Learning Frameworks

- **TensorFlow** – Google's open-source ML framework for large-scale models.
- **Py Torch** – popular for research and production, flexible and developer-friendly.
- **Keras** – high-level API for rapid prototyping.
- **Scikit-learn** – classical ML algorithms (regression, clustering, classification).

AI Coding Assistants

- **GitHub Copilot** – AI-powered codecompletion.
- **Tabnine** – predictive code suggestions.
- **ChatGPT / Claude Code** – naturallanguage coding support.
- **Mutable.ai** – automates documentation andtesting.

Cloud AI Platforms

- **Google Cloud AI** – APIs for vision, NLP, translation, and AutoML.
 - **AWS AI/ML Services** – SageMaker for training/deployment, Rekognition for vision.
- Microsoft Azure AI** – Cognitive Services, ML Studio for scalable AI solutions.

Key Certifications for AI Developers

Foundational ML & AI

- *Machine Learning by Andrew Ng (Coursera)* → core ML concepts.
- *Deep Learning Specialization (DeepLearning.AI)* → advanced neural networks.

NLP & Language Models

- *Coursera NLP Specialization (DeepLearning.AI)* → sentiment analysis, transformers.
- *Hugging Face NLP Course* → hands-on with modern NLP tools.

Cloud & Deployment

- *AWS Certified Machine Learning – Specialty* → training/deployment on AWS.
- *Microsoft Azure AI Engineer Associate* → scalable AI solutions on Azure.
- *Google Cloud ML Engineer* → ML design and deployment on GCP.
- *Docker Essentials Certification* → containerization for AI apps.

Ethics & Responsible AI

- *IBM AI Engineering Professional Certificate* → ML, DL, deployment, ethics.
- *AI Ethics & Responsible AI (edX/Harvard)* → fairness, bias mitigation, transparency.



Project Types



AI Developers can showcase their capabilities through projects in **NLP, computer vision, recommendation systems, and data deployment**, each demonstrating skills in language models, visual data processing, personalization, and production-ready AI. These projects highlight both **technical expertise** (Python, ML frameworks, cloud tools) and **business impact** by improving customer engagement, security, personalization, and scalable automation.

What challenges do you anticipate in pursuing this career path?

- As an AI Developer, key challenges include keeping up with rapidly evolving technologies, managing complex deployments, and ensuring access to high-quality data. The role also demands continuous learning, collaboration across teams, and balancing technical expertise with ethical responsibility while proving real business value in a competitive field.

Challenges of Being an AI Developer

- **Always changing:** The tools and computers we use keep getting new updates, so you have to keep learning.
- **Messy data:** AI needs lots of information, but sometimes that information is wrong or incomplete.
- **Fairness:** AI can accidentally treat people unfairly, so developers must be careful to make it fair.
- **Lots of competition:** Many smart people want to be AI developers, so you have to work hard to stand out.
- **Mix of skills:** It's not just coding — you also need math, problem-solving, and knowledge about different subjects.
- **Big projects:** Making a small AI is easy, but turning it into something millions of people can use is tough.
- **Rules and laws:** Governments are starting to make rules about AI, so developers must follow them.

Ways to Stay Updated

- **Continuous Learning** → Take online courses, certifications, and workshops to keep pace with evolving tools and frameworks.
- **Research & Reading** → Follow AI journals, blogs, and publications to stay informed about breakthroughs and trends.
- **Community Engagement** → Participate in developer forums, open-source projects, and conferences to learn from peers and experts.
- **Hands-On Practice** → Experiment with new libraries, frameworks, and cloud services to apply knowledge in real projects.

10 month roadmap

PHASE 1-3: FOUNDATIONS

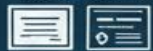


Skills: Python programming, statistics, probability, linear algebra

Tools: Python scripts, data cleaning & visualization, text classification (spam vs ham)



Certifications:



Certifications:

Intro to Python, Mathematics for Machine Learning ML by Andrew Ng

Certification



PHASE 4-6: NLP & DEEP LEARNING



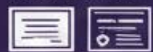
Message: Gain hands-on NLP expertise and apply learning to language tasks

Skills: NLP basics (tokenization, stemming, lemming), deep learning for text

Projects: Sentiment analyzer text generator (poetry/chatbot)



Certification



Certifications: Coursera NLP Specialization, Deep Learning Specialization

Tools: PyTorch, TensorFlow



PHASE 7-10: DEPLOYMENT, ETHICS & PORTFOLIO

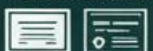


Skills: Model deployment, APIs, cloud MLOps, ethics in NLP, bias detection, LIME, SHAP

Projects: Deploy NLP model as end-to-end ML pipeline, bias detection case study, final portfolio (2-3 projects)



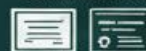
Certifications:



Certifications:

IBM AI Engineering, AWS ML Specialty

Certifications:





Thank you